

Causality

David Puelz

The University of Austin

Cause-and-Effect



Cause-and-effect

Potential outcomes and counterfactuals

Observational studies



These two ideas seem completely unrelated.

But, they are and stealthily show up **EVERYWHERE!**



"Our tariffs on China will bring back American manufacturing and create millions of jobs." —

President Trump, 2025



"Our tariffs on China will bring back American manufacturing and create millions of jobs." —

President Trump, 2025

"Banning seed oils and processed food in school lunches will end the childhood obesity epidemic."

— RFK Jr. / MAHA, 2025



"Our tariffs on China will bring back American manufacturing and create millions of jobs." —

President Trump, 2025

"Banning seed oils and processed food in school lunches will end the childhood obesity epidemic."

— RFK Jr. / MAHA, 2025

"Striking Iran's nuclear sites will set their weapons program back by a decade." — Sen. Lindsey

Graham, 2025



"Our tariffs on China will bring back American manufacturing and create millions of jobs." —

President Trump, 2025

"Banning seed oils and processed food in school lunches will end the childhood obesity epidemic."

— RFK Jr. / MAHA, 2025

"Striking Iran's nuclear sites will set their weapons program back by a decade." — Sen. Lindsey

Graham, 2025

"Newsom's environmental regulations are why businesses are fleeing California for Texas." —

President Trump, 2025



"Our tariffs on China will bring back American manufacturing and create millions of jobs." —

President Trump, 2025

"Banning seed oils and processed food in school lunches will end the childhood obesity epidemic."

— RFK Jr. / MAHA, 2025

"Striking Iran's nuclear sites will set their weapons program back by a decade." — Sen. Lindsey

Graham, 2025

"Newsom's environmental regulations are why businesses are fleeing California for Texas." —

President Trump, 2025

"Mass deportations will raise wages and reduce violent crime in American cities." — Trump

administration, 2025



"Our tariffs on China will bring back American manufacturing and create millions of jobs." —

President Trump, 2025

"Banning seed oils and processed food in school lunches will end the childhood obesity epidemic."

— RFK Jr. / MAHA, 2025

"Striking Iran's nuclear sites will set their weapons program back by a decade." — Sen. Lindsey

Graham, 2025

"Newsom's environmental regulations are why businesses are fleeing California for Texas." —

President Trump, 2025

"Mass deportations will raise wages and reduce violent crime in American cities." — Trump

administration, 2025

"Slashing federal waste through DOGE will lower inflation and reduce the deficit." — Elon Musk,

2025



These two paradigms are one and the same!

Cause is a statement of something being manipulated or changed

Effect is a measure of the change in an outcome of interest



These two paradigms are one and the same!

Cause is a statement of something being manipulated or changed

Effect is a measure of the change in an outcome of interest

“Cause” is the same as a policy introduction or change

“Effect” is the unique, independent measurement of how the cause modulated some other part of our system



We see **causes** all of the time

The federal government increases the minimum wage

The FAA mandates face coverings on planes

The percentage of vaccinated individuals exceeds 50% of population

It's windy on an Austin afternoon

Sweetgreen decreases its Kale caesar salad price by \$2

States implement stay-at-home orders during the pandemic



Causes might also just be variation in a population

Are dog-lovers nicer people?

Does race affect hiring decisions?

Does age affect COVID-19 mortality?

Is there a “gender-gap” in salary?



The **difficult** question is, what about the **effects**?

What outcomes do we look at?

How do we measure them?

Are there other variables that might affect the outcomes *and* the causes?



The **difficult** question is, what about the **effects**?

What outcomes? (**lower class unemployment rate, income, ...**)

How to measure? (**government data, surveys, ...**)

Other variables? (**current economic conditions, differences among states, ...**)



The **difficult** question is, what about the **effects**?

What outcomes? (**revenue, count of kale caesars sold, ...**)

How to measure? (**financial data, ...**)

Other variables? (**seasonality, temperature, weather, ...**)



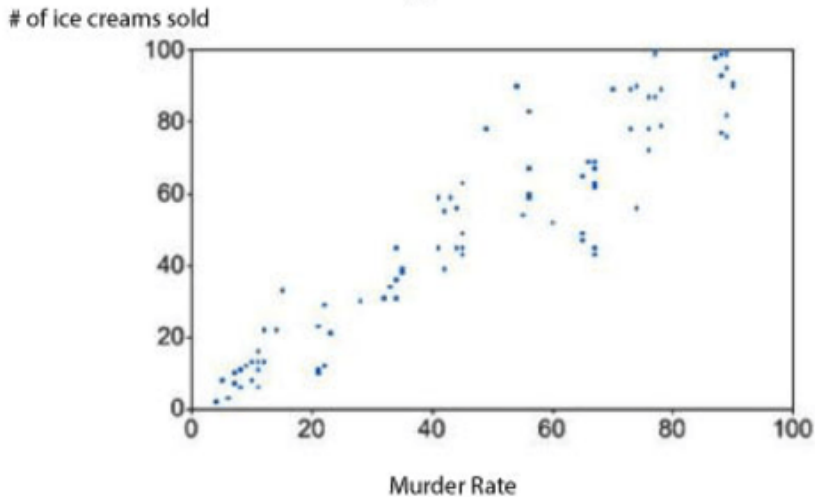
The **difficult** question is, what about the **effects**?

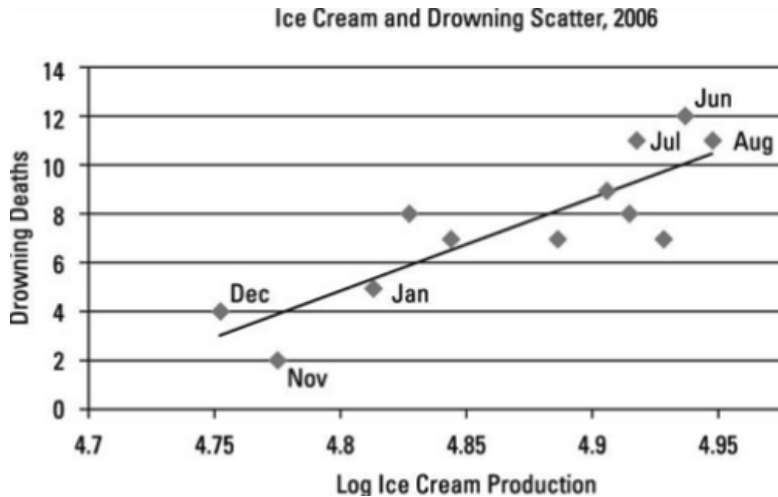
What outcomes? (**whether a job applicant receives a callback**)

How to measure? (**follows directly from above**)

Other variables? (**other resume characteristics, GPA, university, ...**)

Figure 1







Challenges are related to both the **system of study** and **ability to gather the right data**.

With data in hand, you can start to formulate hypotheses and test them.

There might be lurking variables driving an underlying relationship (**ice cream**). Only an expert (you!) can identify those and take them into account.

Let's formalize these ideas with some basic notation

Potential Outcomes and Counterfactuals



Units of study indexed by i (**states, stores, customers, people**)

Cause aka treatment aka policy for unit i : z_i

$$z_i = \begin{cases} 0 & \text{not treated (control)} \\ 1 & \text{treated} \end{cases}$$

Outcome for each i as Y_i , written as a function of the **cause**:

$$Y_i(z_i) \leftarrow \text{potential outcome for unit } i$$

With this notation, we have the building blocks to talk about policy **effects**!



How did state lockdowns at the beginning of 2020 affect the spread of the virus?

What are the **potential outcomes**?



Let's set up the structure of this problem. Let i denote a state:

$$i \in \{\text{New York, California, Florida, Texas, South Dakota, } \dots\}$$

First, we define z_i :

$$z_i = \begin{cases} 0 & \text{no lockdown} \\ 1 & \text{lockdown} \end{cases}$$



Let's set up the structure of this problem. Let i denote a state:

$$i \in \{\text{New York, California, Florida, Texas, South Dakota, } \dots\}$$

First, we define z_i :

$$z_i = \begin{cases} 0 & \text{no lockdown} \\ 1 & \text{lockdown} \end{cases}$$

Second, we define our outcome: Y_i = cases per capita (in state i) after lockdown or no lockdown.



A brief aside: Defining exactly what the treatment z_i is very hard! It could be a combination of many factors.

Masking

Bar and restaurant closures

School closures

Curfews

Limits to exercise

Retail store closures

Q: How would you define z_i ?



i (state)	z_i (lockdown)	$Y_i(0)$	$Y_i(1)$
New York			
Florida			
California			
Texas			
South Dakota			
Illinois			
$\vdots$	$\vdots$	$\vdots$	$\vdots$



i (state)	z_i (lockdown)	$Y_i(0)$	$Y_i(1)$
New York	1		
Florida	0		
California	1		
Texas	0		
South Dakota	0		
Illinois	1		
$\vdots$	$\vdots$	$\vdots$	$\vdots$



i (state)	z_i (lockdown)	$Y_i(0)$	$Y_i(1)$
New York	1		.0034
Florida	0	.007	
California	1		.0014
Texas	0	.004	
South Dakota	0	.0028	
Illinois	1		.002
$\vdots$	$\vdots$	$\vdots$	$\vdots$



i (state)	z_i (lockdown)	$Y_i(0)$	$Y_i(1)$
New York	1	??	.0034
Florida	0	.007	??
California	1	??	.0014
Texas	0	.004	??
South Dakota	0	.0028	??
Illinois	1	??	.002
$\vdots$	$\vdots$	$\vdots$	$\vdots$

We are able to know both of the potential outcomes for each state!



We can define the **causal effect** of the “lockdown treatment” as the difference between the two potential outcomes.

$$\tau_{NY} = Y_{NY} \left(\text{lockdown} \right) - Y_{NY} \left(\text{no lockdown} \right)$$

Or written more generally:

$$\tau_i = Y_i(1) - Y_i(0)$$



We only observe one of the two potential outcomes for each unit.

Economics of COVID policy: a state either locks down or doesn't

Drug trials: an individual either receives the medicine or the placebo

Gender wage gap: a person is either male or female

The unknown outcomes are called the **missing potential outcomes** or **counterfactuals**.
This is what makes causality a nontrivial task... it is a [missing data problem](#).



The potential outcomes framework is **always** our starting point.

Note: Randomized controlled trials (RCTs) are the **gold standard**—when treatment is randomly assigned, treatment and control groups are comparable, and causal effects are straightforward to estimate.



The potential outcomes framework is **always** our starting point.

Note: Randomized controlled trials (RCTs) are the **gold standard**—when treatment is randomly assigned, treatment and control groups are comparable, and causal effects are straightforward to estimate.

But in many settings we have **observational data**: treatment is *not* randomized, it is simply **observed**.

“I observe states lockdown and not lockdown”

“I observe minimum wage increases in some places and not others”

We can't control what we observe, but we can still try to use this variation to answer interesting causal questions!

Observational Studies



Let's use the minimum wage increase in NJ as our running example.

1992, NJ increased minimum wage from \$4.25 to \$5.05 per hour

Causal question: Did this increase **reduce** or **increase** employment?

Several approaches:

- Compare outcome in NJ to PA: **cross-sectional comparison**

- Compare similar groups in NJ and PA: **subclassification / statistical control**

- Compare same group before and after: **panel comparison / before-after**

- Compare before-after in NJ to before-after in PA: **difference-in-differences**



Define $\bar{Y}^{NJ}$ and $\bar{Y}^{PA}$ as average % full-time employees in fast-food restaurants.

$$\text{causal effect} = \bar{Y}^{NJ} (\text{treatment}) - \bar{Y}^{PA} (\text{control})$$

Q: What are some issues?



Define $\bar{Y}^{NJ}$ and $\bar{Y}^{PA}$ as average % full-time employees in fast-food restaurants.

$$\text{causal effect} = \bar{Y}^{NJ} (\text{treatment}) - \bar{Y}^{PA} (\text{control})$$

Q: What are some issues?

Confounding! NJ and PA might differ in economies, politics, populations, etc.



Pick covariates X that might be confounding. Compute the causal effect only on units with **identical** values of X .

$$\text{causal effect} = \bar{Y}_{X=\text{Taco Bell}}^{\text{NJ}} - \bar{Y}_{X=\text{Taco Bell}}^{\text{PA}}$$

Matching: compare apples-to-apples!

Q: What other X 's could be used?



Pick covariates X that might be confounding. Compute the causal effect only on units with **identical** values of X .

$$\text{causal effect} = \bar{Y}_{X=\text{Taco Bell}}^{\text{NJ}} - \bar{Y}_{X=\text{Taco Bell}}^{\text{PA}}$$

Matching: compare apples-to-apples!

Q: What other X 's could be used?

Geographic location, closeness to NJ-PA border, urban or rural area



What if we have data measured *over time*? Panel or longitudinal data.

$$Y_{i,\text{before}}^{\text{NJ}} \longrightarrow \text{min wage hike} \longrightarrow Y_{i,\text{after}}^{\text{NJ}}$$

$$Y_{i,\text{before}}^{\text{PA}} \longrightarrow \text{min wage hike} \longrightarrow Y_{i,\text{after}}^{\text{PA}}$$

$$\text{causal effect} = \bar{Y}_{\text{after}}^{\text{NJ}} - \bar{Y}_{\text{before}}^{\text{NJ}}$$

Q: What are some issues?



What if we have data measured *over time*? Panel or longitudinal data.

$$Y_{i,\text{before}}^{\text{NJ}} \longrightarrow \text{min wage hike} \longrightarrow Y_{i,\text{after}}^{\text{NJ}}$$

$$Y_{i,\text{before}}^{\text{PA}} \longrightarrow \text{min wage hike} \longrightarrow Y_{i,\text{after}}^{\text{PA}}$$

$$\text{causal effect} = \bar{Y}_{\text{after}}^{\text{NJ}} - \bar{Y}_{\text{before}}^{\text{NJ}}$$

Q: What are some issues?

Confounding again! Time is potentially an issue (remember ice cream).

Remove time-trend confounding by using panel data from both **treated** (NJ) and **control** (PA).

Four outcomes: $\bar{Y}_{\text{before}}^{\text{NJ}}$, $\bar{Y}_{\text{after}}^{\text{NJ}}$, $\bar{Y}_{\text{before}}^{\text{PA}}$, $\bar{Y}_{\text{after}}^{\text{PA}}$

$$\text{DiD} = (\bar{Y}_{\text{after}}^{\text{NJ}} - \bar{Y}_{\text{before}}^{\text{NJ}}) - (\bar{Y}_{\text{after}}^{\text{PA}} - \bar{Y}_{\text{before}}^{\text{PA}})$$

Why does this work?

answer 1: **picture**

answer 2: **some math**



$4 \times N$ potential outcomes, 2 for each state: $\{Y_i^{\text{NJ}}(0), Y_i^{\text{NJ}}(1), Y_i^{\text{PA}}(0), Y_i^{\text{PA}}(1)\}$ for all i

$$\begin{aligned}\text{DiD} &= (\bar{Y}_{\text{after}}^{\text{NJ}} - \bar{Y}_{\text{before}}^{\text{NJ}}) - (\bar{Y}_{\text{after}}^{\text{PA}} - \bar{Y}_{\text{before}}^{\text{PA}}) \\ &= \left(\mathbb{E}[Y_i^{\text{NJ}}(1)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{before}] \right) \\ &\quad - \left(\mathbb{E}[Y_i^{\text{PA}}(0)|\text{after}] - \mathbb{E}[Y_i^{\text{PA}}(0)|\text{before}] \right) + 0 \\ &= \left(\mathbb{E}[Y_i^{\text{NJ}}(1)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}] \right) \\ &\quad + \left(\mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{before}] \right) \\ &\quad - \left(\mathbb{E}[Y_i^{\text{PA}}(0)|\text{after}] - \mathbb{E}[Y_i^{\text{PA}}(0)|\text{before}] \right)\end{aligned}$$



$$\begin{aligned}\text{DiD} = & \left(\mathbb{E}[Y_i^{\text{NJ}}(1)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}] \right) \\ & + \left(\mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{before}] \right) \\ & - \left(\mathbb{E}[Y_i^{\text{PA}}(0)|\text{after}] - \mathbb{E}[Y_i^{\text{PA}}(0)|\text{before}] \right)\end{aligned}$$

First part: $\mathbb{E}[Y_i^{\text{NJ}}(1)|\text{after}] - \mathbb{E}[Y_i^{\text{NJ}}(0)|\text{after}]$ = sample average treatment effect (of treated)

Second part: (slope of NJ line) – (slope of PA line)

Parallel trend assumption: second part = 0



The DiD estimate can be obtained from a simple **regression**. Suppose we have panel data with units i observed at times t :

$$Y_{it} = \alpha + \beta_1 \cdot \text{Treat}_i + \beta_2 \cdot \text{Post}_t + \delta \cdot (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it}$$

$\text{Treat}_i = 1$ if unit i is in the treatment group (e.g., NJ)

$\text{Post}_t = 1$ if time t is after the intervention

δ is the **DiD estimator** — this is the causal effect!